spatiAlert

Getting Started Guide

spatiAlert identifies geographic hotspots of undervaccination using the Getis-Ord Gi* spatial statistic. This guide walks through data setup, geography and spatial-weights choices, exploring your data, and how to interpret your results — so you can go from a spreadsheet to a hotspot map with confidence.

> This app accompanies the analysis published as a research letter in the New England Journal of Medicine:
>
> *Clusters of Concern — Spatial Link between Childhood Undervaccination and Measles Outbreaks in South Carolina.* Published May 20, 2026. N Engl J Med 2026;394:2479–2481. DOI: 10.1056/NEJMc2604004
>
> Full text: https://www.nejm.org/doi/full/10.1056/NEJMc2604004

Switch on **"Use defaults"** at the top of the Analysis tab (or click **"Run default analysis"**) to start from the recommended settings; see Section 7.

1. Setting Up

After unzipping the spatiAlert download, you'll have a folder structure like this:

```
spatialert (your chosen folder)
  |-- run_spatialert - user.R  <- open and run this file
  |-- spatiAlert_Getting_Started.pdf  <- this guide
  |-- spatialert_package/
      |-- spatialert/ (the app itself — don't rename or move this)
```

To launch the app, open **run_spatialert - user.R** in RStudio and set the `dir` variable near the top of the file to the folder that *directly contains* `spatialert_package` — not `spatialert_package` itself, and not the folder inside it.

Example: if you unzipped the download to your Desktop, so that the full path to the package looks like: `/Users/JohnDoe/Desktop/spatialert/spatialert_package/spatialert`
then you would set:

```
dir <- "/Users/JohnDoe/Desktop/spatialert"
```

Then click Source (or Ctrl+Shift+S / Cmd+Shift+S). If the path is wrong, the app will tell you specifically what looks off (e.g. pointed too deep, or the folder doesn't exist) rather than a generic error.

2. The Workflow

spatiAlert is organized into tabs, used roughly in order. All analysis runs locally on your machine — nothing is uploaded to any server.

Tab	What you do
Data & Geography	Upload your data, choose what to analyze, and load or upload a geography (tracts, counties, ZIPs, or your own boundary file).
Explore	Look at your data before analyzing it: choropleth and point maps, a ranked table, a data summary, and printable maps.

Tab	What you do
Analysis	Choose spatial weights, decide how areas with no schools are handled, run the Gi* hotspot statistic (or use the defaults), and view the resulting hotspot map.
Results & Export	View the results tables (including schools of concern), export a CSV, generate a Word report, and copy ready-made methods text.
FAQ / Help	Plain-language answers to common questions, including how to read a hotspot.

3. What Data Can spatiAlert Handle?

spatiAlert accepts two data structures. Both are CSV or Excel files.

	Mode A — School / facility level	Mode B — Pre-aggregated
One row per...	school (or school–grade combination) or facility	geographic unit (tract, county, ZIP, or your own areas)
Requires	Latitude & longitude columns for each facility	A geographic ID column that matches your geography
What the app does	Spatially joins each facility to the geography you load, then sums or aggregates values within each area	Matches your rows directly to the geography using the ID column — no spatial join needed
Best when...	You have facility coordinates but no pre-existing geographic assignment	Your source data (e.g. a state health department file) already reports counts or rates by area

Mode A details

- Two separate numeric columns, e.g. latitude and longitude (decimal degrees, e.g. 34.0007, -81.0348). spatiAlert looks for columns containing “lat” and “lon”/“lng” automatically, but you can remap them.
- Address-only data (no lat/lon) is not automatically geocoded — geocode addresses to lat/lon first (e.g. in Excel, ArcGIS, or a geocoding API) before uploading.
- Rows with missing coordinates are dropped, with a warning showing how many.
- **Optional school name and grade columns.** If your file has them, the app can label schools on maps and in the schools table. If a school appears on more than one row (one row per school per grade), the rows are combined: enrollment is summed and the vaccination rate is enrollment-weighted.
- **Messy values are handled.** Percent strings (e.g. “92%”), blanks, and small-cell suppression marks such as “<10” are recognized and treated sensibly; the app reports how many rows were affected.

Mode B details

Geography	ID format accepted	Example
Census tract	11-digit GEOID (2-digit state + 3-digit county + 6-digit tract)	45079010200
County	5-digit FIPS (2-digit state + 3-digit county)	45079
ZIP / ZCTA	5-digit ZIP Code Tabulation Area	29201
Your own areas	Any ID column that is also in your boundary file	e.g. a school district code

IDs are matched as text, so leading zeros matter — make sure Excel hasn't stripped a leading zero from a FIPS/GEOID column (a common cause of unmatched rows).

Using your own geography

Besides the built-in tract, county, and ZIP layers, you can upload a boundary file: a **.zip** containing a shapefile, the individual shapefile parts (.shp, .shx, .dbf, .prj) selected together, or a **GeoJSON** file. After upload:

- Pick the **ID column** from the dropdown (for Mode B, this must match the ID column in your data).
- Tell the app **what the areas represent** (e.g. “school districts”). That wording is used in the maps, tables, and report text.
- Check ID formats: if your data has “1234” but the boundary file has “01234”, they will not match. Partial matches are allowed but the app warns you with the count.

4. What Do You Want to Analyze?

Before mapping columns, spatiAlert asks what variable the hotspot analysis should actually run on. This choice is independent of how your data happens to be recorded — you can have percent-formatted data and still choose to analyze the underlying count, or vice versa.

Target	What it highlights	Typically used for
Count of undervaccinated students	Absolute burden — areas with the most undervaccinated students in raw numbers	Resource allocation, outreach targeting (this is what the published NEJM analysis used)
Undervaccination rate	Relative risk — areas with the highest proportion undervaccinated, regardless of population size	Comparing small and large areas fairly; equity analyses
Vaccination rate	The mirror image of undervaccination rate — hotspots are areas of unusually *high* coverage, and the areas of concern are the coldspots	Identifying high-performing areas or benchmarking

Whichever you choose, you'll then tell the app how that value is recorded in your file: either as a column you already have directly, or calculated by the app from an eligible-population column plus a vaccinated count or rate. As you map columns, the app shows a live table of exactly what will be calculated using your actual selected column names, plus a “You are analyzing: ___” confirmation banner — so it's unambiguous what the hotspot map represents before and after you run the analysis.

5. Exploring Your Data

The Explore tab lets you look at your data before running any statistics. Use the **Go to Explore** box at the bottom of Data & Geography, and the **Go to Analysis** buttons at the top and bottom of the Explore sidebar when you are ready to move on.

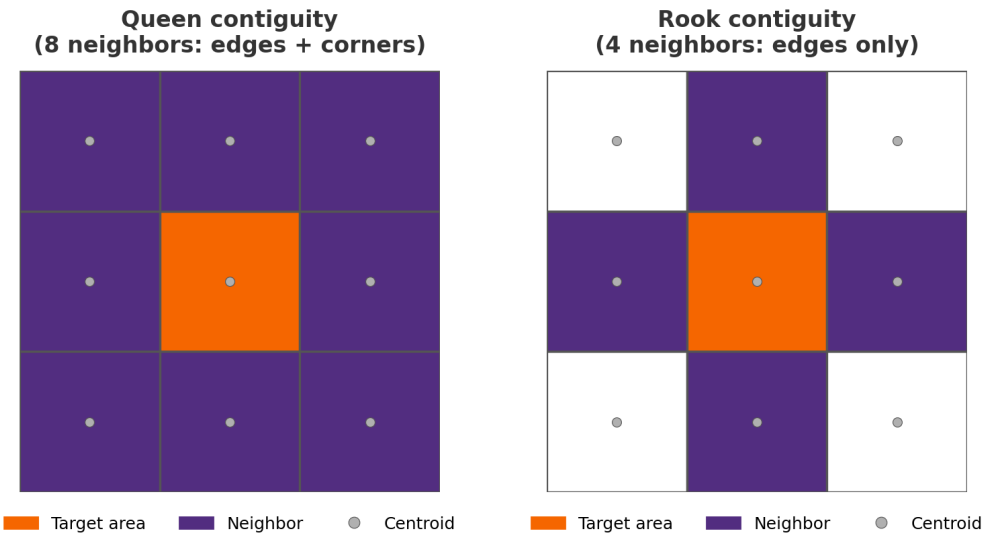
- **Map types.** Choropleth maps (areas shaded by value) and point maps (individual schools) can be shown on an interactive map.
- **Color classes.** Choose how values are grouped into colors: *quantiles* (3–8 classes), *round-number breaks*, *natural breaks (Jenks)*, *equal intervals*, or *custom thresholds* that you type in (remembered separately for each measure). If one top class stretches over a very wide range (for example 44 to 1,487), try round-number or natural breaks. Vaccination rate keeps fixed, meaningful bins (<85, 85–89.9, 90–94.9, ≥95).

- **Ranked table and data summary.** They appear as tabs beside the map. Selecting one or more rows in the ranked table isolates those areas (or schools) on the map; use **Show all on map** to reset.
- **Printable maps.** Create maps as Word, PDF, or PNG files: one map per page or a 2×2 grid, portrait or landscape, with captions and a scale bar below the map. Printed maps do not use online map tiles, so they work offline.

6. Choosing Spatial Weights

Spatial weights define which areas count as each other's “neighbors” — this determines what gets compared to what in the hotspot statistic.

Weights style	How neighbors are defined	When to use it
Queen contiguity	Areas sharing any border point (edge or single vertex)	Simple, standard choice when the geography tiles cleanly with no gaps or islands
Rook contiguity	Areas sharing a border edge (not just a corner)	More conservative version of queen; same caveat about needing full connectivity
K-nearest neighbors (KNN)	Each area's k closest centroids, regardless of shared borders	Guarantees every area has neighbors — essential when geography has islands, non-contiguous areas, or big gaps (e.g. ZIP codes, some tract layouts)





Note how the KNN diagram uses the same filled-polygon style as queen/rook above, but the purple “neighbor” areas are picked by distance between centroids (dashed lines), not by touching the target area — a KNN neighbor can be non-adjacent, while an adjacent area can be excluded if a closer one exists elsewhere.

Checking whether you can use queen/rook

On the Analysis tab, once your geography loads, spatiAlert automatically shows an FYI panel with the mean, median, and range of neighbor counts under both queen and rook contiguity, and flags any areas with zero neighbors. If either shows 0-neighbor areas, those areas would be silently excluded from a queen/rook analysis — use KNN instead to keep every area in the analysis.

Why KNN consensus is the default

A single choice of k (e.g. “k=8”) is somewhat arbitrary — results can shift depending on exactly how many neighbors you use. spatiAlert's default mode instead runs Gi* separately across several k values (4, 6, 8, 10, and 12 by default) and calls an area a robust hotspot only if it is significant in at least a minimum number of those specifications (2 of 5 by default). This is a robustness check across neighborhood-size assumptions, not a single arbitrary pick.

Binary vs. row-standardized weights

Coding	What it means	Recommended for
Binary (B)	Each neighbor counts equally (weight = 1)	Counts (e.g. number of undervaccinated students) — preserves the additive nature of counts
Row-standardized (W)	Each area's neighbor weights sum to 1 (so more neighbors = smaller individual weight)	Rates or proportions — avoids areas with many neighbors being disproportionately influential

7. The Statistics

Local hotspots: Getis-Ord Gi*

spatiAlert uses the Getis-Ord Gi* statistic. For each area, Gi* compares the values in that area *and its neighbors* to the study area as a whole, producing a z-score: positive and statistically significant means a hotspot (values are unusually high in that local area); negative and significant means a coldspot. The “star” matters: **Gi* includes the area itself** in its own neighborhood (plain Gi leaves it out), so an area with a very high value can be flagged on the strength of itself plus its neighbors.

Because a hotspot reflects an area *and its neighbors*, an area with the single highest count is not guaranteed to be a hotspot if its surroundings are low, and an area with modest values can be a hotspot if it sits among high ones. Hotspots point to where undervaccinated students are statistically significantly clustered, not simply where the largest numbers are.

Global G test: is there clustering across the whole area?

Alongside the local results, the app runs the **Global G** test, which asks a single question: across the entire study area, are high values clustered together more than chance would produce? It uses the **same weights** as your analysis (for KNN consensus runs, the middle k), binary coding, and a one-sided test for clustering of high values. A small p-value means there is overall clustering, which supports looking at the local hotspots; a large p-value means the overall pattern is not distinguishable from random. It is shown on the Results tab and in the report. It needs non-negative values, and for rates the app notes that binary weights are used for this test.

Multiple comparisons correction

The app's default is **None**. You can choose a correction on the Analysis tab:

Option	What it does	When to use
None (default)	No correction — more areas flagged, some may be false positives	Exploratory identification of candidate areas, as in the published NEJM analysis
False discovery rate (FDR)	Controls the expected proportion of false positives among significant results	A good choice when you want fewer false positives but still exploratory use
Bonferroni	Very conservative; divides the significance threshold by the number of tests	When you need strict control of false positives, e.g. confirmatory analyses

Areas with no schools

When you map schools to areas, some areas may have no schools at all. The Analysis tab lets you choose how they are treated:

Option	What it does
Count as zero	The area is kept and given a value of zero. Default when analyzing counts, since no schools means no undervaccinated students.
Exclude from the analysis	The area is dropped and is not used as a neighbor. Default for rates, which are undefined with no students. Excluded areas appear in light gray on the hotspot map.
Use the study-area average	The area is kept and given the average value of areas that do have schools.

The choice you make is recorded in the methods text and report. An area with no schools can still be flagged as a hotspot because its neighbors have high values; see the app FAQ for how to read that.

Default settings

Switch on **“Use defaults”** at the top of the Analysis tab, or click **“Run default analysis”**, to apply:

- Weights: K-nearest neighbors, consensus across k = 4, 6, 8, 10, 12 (binary coding)
- Consensus rule: hotspot/coldspot requires significance in ≥2 of the 5 specifications
- Statistic: Getis-Ord Gi*
- Correction: none
- Significance threshold: $\alpha = 0.05$

While the switch is on, these settings are locked so nothing is changed accidentally — toggle it off to make your own selections. The analysis variable is set separately on the Data & Geography tab (the published analysis used the count of undervaccinated students).

8. Results & Export

- **Hotspot map (Analysis tab).** After you run the analysis, hotspots and coldspots are shown on the map on the Analysis tab. A schools layer can be turned on over this map. For the vaccination-rate target, hotspots are areas of high coverage and the areas of concern are coldspots.
- **Results tables (Results & Export tab).** The ranked table of areas and the schools table are on the Results & Export tab.
- **Schools of concern.** Schools located in areas of concern are listed with a color key. The default threshold is 95% (adjustable), and schools with several grades are combined into one row.
- **Word report.** Includes a plain-language “About this analysis” section, an explanation of the Global G result, footnoted tables, and methods text that adapts to the choices you made (weights, correction, no-school handling).
- **CSV export and methods text.** Download results as CSV or copy ready-made methods text for your own write-up.

9. Troubleshooting

The app won't launch / “could not find the app” error

Check the `dir` path set at the top of run_spatialert - user.R — it should point to the folder that directly contains `spatialert_package`, not the package folder itself. See Section 1 for a worked example.

I selected “percent vaccinated” — what is the hotspot analysis actually run on?

Whatever you selected under “What do you want to analyze?” — not automatically the count. The app always shows a “You are analyzing: ___” banner on the Data & Geography and Analysis tabs, plus a live calculation table showing your actual column names.

My uploaded boundary file doesn't match my data

Check that the ID column you chose is the same kind of ID as in your data (same format, same number of digits, leading zeros intact). If only some IDs match, the app warns you with the count of unmatched rows — fix those IDs and re-upload.

The map background is blank

Interactive maps use online map tiles, which need an internet connection. Offline, your areas and points still draw, just without a background; printed maps never use tiles.

My geography has areas with no neighbors

Switch to KNN weights on the Analysis tab, which guarantee every area has neighbors. The connectivity FYI panel tells you whether queen/rook is viable.

My state only has a few counties. Will Gi* work?

Gi* has limited statistical power with fewer than ~20 units. Consider tract-level data, or interpret county-level results cautiously as a screening exercise.

Column mapping looks wrong / I can't tell what's being calculated

The live “Exactly what will be calculated” panel on the Data & Geography tab shows your column names plugged into the formula — check it before confirming.

Where can I get help beyond this guide?

See the in-app FAQ and Help tabs, or reach out to the app's maintainers directly.

10. References

For readers who want to go deeper on the methodology used in this app:

Clusters of Concern — Spatial Link between Childhood Undervaccination and Measles Outbreaks in South Carolina. Published May 20, 2026. N Engl J Med 2026;394:2479–2481. DOI: 10.1056/NEJMc2604004 — the published analysis this app accompanies.

Getis, A. and Ord, J.K. (1992). The Analysis of Spatial Association by Use of Distance Statistics. *Geographical Analysis*, 24(3), 189–206. — the original Gi/Gi* statistic and the global G statistic.

Ord, J.K. and Getis, A. (1995). Local Spatial Autocorrelation Statistics: Distributional Issues and an Application. *Geographical Analysis*, 27(4), 286–306. — distributional properties and inference for Gi*.

Cliff, A.D. and Ord, J.K. (1981). *Spatial Processes: Models & Applications.* Pion, London. — foundational reference for spatial weights matrices.

Benjamini, Y. and Hochberg, Y. (1995). Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing. *Journal of the Royal Statistical Society: Series B*, 57(1), 289–300. — basis for the FDR correction offered in this app.